

Characterization of Polymer Molecular Weight Distribution by NMR Diffusometry: Experimental Criteria and Findings

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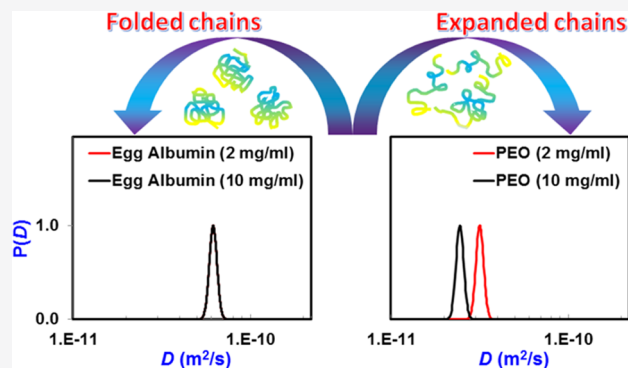


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ABSTRACT: NMR diffusometry finds useful applications in characterizing molecular weight (M) and molecular weight distribution (MWD) for polymers due to its unique advantages in generic detection, chemical selectivity, and quantitation. Here, we present a fundamental study to explore how the condition of diffusion measurement impacts the determined MWD. We use the critical dilute concentration C_{dilute}^* to explicitly delineate the boundary of the sufficiently dilute condition, below which chain interactions have a negligible impact on polymer diffusion. We present solid evidence to validate the postulated theory that links C_{dilute}^* to molecular weight, polydispersity, and chain conformation. Quantitative analysis reveals the consequence of violating the sufficiently dilute condition with M /MWD characterization. These findings provide useful guidance for M /MWD characterization by NMR diffusion and help to rationalize the data disparity that exists in the literature. The results further provide new insights into the interplay between chain conformations and diffusion for globular structure, such as proteins, and provide a different approach toward characterizing polymer architecture and molecular weight.



INTRODUCTION

Polymers undoubtedly represent one of the most important classes of functional materials that play an indispensable role in modern industry and daily life. Analysis of molecular weight (M) and molecular weight distribution (MWD) is crucial to direct polymer synthesis and processing since this information fundamentally impacts the properties and performance of polymeric materials in many aspects, ranging from physical properties to mechanical properties to chemical resistance.^{1–3} Though gel permeation chromatography (GPC) is a well-known technology suited for M /MWD characterization, NMR diffusometry continues to find expanding applications in M /MWD characterization^{4–6} due to its uniquely combined advantages in generic detection, chemical selectivity, non-destructiveness, and quantitative nature. The technique⁷ utilizes one or several pairs of transient magnetic field gradient to couple the detected NMR signal with the average molecular displacement, allowing determination of the diffusion coefficient associated with individual NMR peaks to quantify M /MWD meanwhile preserving the spectroscopic resolution. This renders NMR some superiority to other techniques, such as GPC and mass spectrometer (MS) in polymer characterization, which often are lacking some of these important features. Easy sample preparation, flexible choice of solvent, and a broad range of operating temperature further strengthen the practicality of diffusion measurement on a routine basis. As research and development progresses, these advantages of NMR diffusometry have become overwhelmingly important

when studying polymer systems with increased structural complexity,^{5,6,8,9} which often necessitates more detailed characterization. Under the dilute condition, polymer diffusion (D) is related to M based on the following equation⁴

$$D = kM^{-\alpha} \quad (1)$$

α is the Mark–Houwink coefficient and k is a constant that depends on the polymer system. By utilizing a set of monodisperse polymer standards to determine k and α to establish the correlation between D and M ,^{4,10} generic determination of the apparent M /MWD of unknown polymeric materials becomes viable.

The past decades have witnessed numerous research efforts using NMR diffusion to investigate polymer systems from different aspects, including chain conformations,^{11–13} polymer size,¹¹ M and MWD of polymers and peptides,^{5,9,10,14–17} and the development of new methods to determine M /MWD.^{4,15,18–22} However, close examination reveals that data disparity often exists when comparing literature results. For example, the pioneering work by Chen et al. and Håkansson et

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al. has laid out the foundation of M/MWD characterization using NMR diffusion.^{4,10} Both groups determined the $D = kM^{-\alpha}$ relationship using monodisperse poly(ethylene oxide) (PEO) in D_2O . Chen et al. reported a value of $k = 2.40 \times 10^{-8}$ (e.g., $10^{-7.62}$) and $\alpha = 0.62$ using standards with M in the range from 1 to 500k. In contrast, Håkansson et al. reported the value of $k = 1.0 \times 10^{-8}$ and $\alpha = 0.55$ using standards with M in the range from ~ 1 to 900k. Though both groups claimed that the measurement was performed under sufficiently dilute conditions, where polymer chain interactions are negligible, the drastic difference between their results undoubtedly signifies the presence of experimental inconsistency. Interestingly, such discrepancy is hardly mentioned or discussed in the literature. It also leaves a question mark regarding what values of k and α (for PEO) should be used for reliable analysis of M/MWD . In another example, Arrabal-Campos et al. and Li et al. have both conducted NMR diffusion on monodisperse polystyrene (PS) in benzene at 21 and 25 °C, respectively.^{14,15} However, their results showed a significant difference in the α value (0.568 vs 0.537) for the same system, which cannot be simply explained by the minor difference in temperature. Diffusion of PS in chloroform has also been extensively studied by NMR and used as a standard for M/MWD analysis, yet a wide range of α values have been reported.^{5,12,21,22} These data disparities, if not treated with care, will translate into undesired systematic deviations in the determined M/MWD and it further brings up some fundamental questions: what causes the disparity and how to resolve the problem?

Unlike small molecules, polymer chains tend to swell in a good solvent and interact with each other to restrict chain motion. Such interactions become further exacerbated as concentration increases and consequently lead to a reduction in polymer diffusion (D_{polymer}). It was noted in both Chen and Håkansson's work that D_{polymer} should be measured under the "sufficiently" dilute condition where polymer chain interactions have a negligible impact. Otherwise, the determined M/MWD will be inflated due to the reduced D_{polymer} . Though the critical overlap concentration C^* is widely used to characterize the dilute condition,²³ it is often not clear how far below C^* can be reliably treated as sufficiently dilute in the context of D_{polymer} measurement. In fact, a common problem associated with M/MWD characterization using D_{polymer} stems from the selection of the aforementioned sufficiently dilute condition. A seemingly low concentration, typically between 0.5 and 4.5 mg/mL, has been used in the majority of polymer diffusion studies for M/MWD characterization,^{4–6,8–12,14–22} yet solid justification is rarely provided. In some extreme cases, the polymer concentration is even up to ~ 80 mg/mL.¹³ From the perspective of a general audience, a more practical question arises: what is the reliability to leverage the literature results for M/MWD characterization?

In this report, we present a fundamental study on D_{polymer} in a solution with the aim to address the above questions. Our goal is twofold: establish a standard protocol to ensure consistent and reliable M/MWD characterization by NMR diffusion and present a new analytical capability that allows characterizing polymer architecture and globular structure through diffusion analysis. To facilitate illustration, the rest of the paper will be organized as follows: first, we introduce the concept of critical dilute concentration C_{dilute}^* to define the sufficiently dilute condition for D_{polymer} measurement. We then present a theory based on the scaling law to show how C_{dilute}^* is connected to relevant structural characteristics of polymer

chains. Second, we design and conduct robust diffusion measurement to validate the postulated theory that captures new insights regarding polymer chain interactions in a solution. Finally, we demonstrate the importance of our validated theory in applications of polymer characterization and will briefly discuss the impact on M/MWD analysis if D_{polymer} is not measured under sufficiently dilute condition. These findings not only establish guidance for M/MWD characterization by NMR diffusion but also enable technical advancement, particularly in studying polymer architecture and globular structure using NMR spectroscopy and diffusometry.

THEORY

Consider a monodisperse and linear flexible polymer system (with molecular weight M) and let V_p and V_s be the total volume pervaded by polymer chain and solvent, respectively. Their ratio f_v denotes the polymer volume fraction,²⁴ provided that $V_p \ll V_s$.

$$f_v = \frac{V_p}{V_s} = \frac{N_A \left(\frac{m_p}{M} \right) v_p}{V_s} \propto \frac{N_A \left(\frac{m_p}{M} \right) R_h^3}{V_s} \quad (2)$$

N_A is the Avogadro constant, m_p is the polymer mass, v_p is the pervaded volume of the individual polymer chain, and R_h is the hydrodynamic radii. $f_{v,\text{dilute}}^*$ and C_{dilute}^* are defined as the respective critical dilute volume fraction and critical dilute mass concentration that explicitly specify what concentration (or below) is sufficiently dilute for reliable diffusion measurements. We consider the change of D_{polymer} as a direct consequence of interchain interactions induced by volume expansion. Thus, the sufficient dilute condition is satisfied at or below C_{dilute}^* (or $f_{v,\text{dilute}}^*$) and $\frac{dD_{\text{polymer}}}{dC} = 0$. Using the scaling law $R_h \sim M^\alpha$, the following equation holds

$$f_{v,\text{dilute}}^* \propto N_A C_{\text{dilute}}^* M^{3\alpha-1} \quad (3)$$

$C_{\text{dilute}}^* = \left(\frac{m_p}{V_s} \right)_{\text{dilute}}^*$ describes the chain conformation in a solution and it equals 0.6 for a random coil, 0.5 for an unperturbed chain, and 1/3 for globular structure. The theory requires $f_{v,\text{dilute}}^*$ as a constant for a given temperature and the polymer/solvent system since it quantitatively characterizes the onset of interactions among polymer chains, which should be dictated by these factors. Equation 4 therefore should hold

$$C_{\text{dilute}}^* = C_0 M^{1-3\alpha} \quad (4)$$

where C_0 is a constant. Equation 4 is a reminiscence of the classical relationship that describes C^* and M .^{23,25} If the above equations hold true, C_{dilute}^* should scale with M via the power law unless polymer chains adopt the globular structure (i.e., $\alpha = 1/3$). In the following, we will examine the validity of the above equations and explore their applications in polymer characterization.

EXPERIMENTAL SECTION

Poly(methyl methacrylate) (PMMA) and poly(ethylene oxide) (PEO) standards with a low polydispersity index (PDI) were purchased from Polymer Laboratories and used as received. Detailed molecular weight information is listed in Table 1. In consideration of $M_n \approx M_w \approx M_p$ for monodispersed samples (PDI ~ 1), peak molecular weight M_p provided by the vendor was consistently used to establish the correlation between

Table 1. Molecular Weight Information of Monodispersed PEO and PMMA Standards^a

sample ID	M_p (kg/mol)	PDI	M_p (kg/mol)	PDI
PMMA	4.9	1.10	174	1.04
PMMA	12.7	1.03	519	1.03
PMMA	28.8	1.03	772	1.03
PMMA	55.9	1.03	1520	1.08
PEO	4.1	1.05	8.5	1.02
PEO	18.3	1.02	32.5	1.02
PEO	114	1.04		

^a M_p is the peak molecular weight and PDI is the polydispersity index.

polymer diffusion (D) and molecular weight (M). Polystyrene 1683 with a broad molecular weight distribution was obtained from Dow Inc. and used as a “standard” for comparison of M /MWD determined by NMR and GPC. Albumin from egg white protein was purchased from Sigma-Aldrich and used as received without further purification.

For GPC analysis, the sample was prepared in a HPLC grade tetrahydrofuran at a concentration of 2 mg/mL and allowed to dissolve for 2 h on a flatbed shaker. The standards were run on a Waters Model e2695 at a nominal flow rate of 1.0 mL/min of HPLC grade tetrahydrofuran. Two 5 μ m mixed-C columns from Polymer Laboratories were used and held at 35 °C with an injection volume of 100 μ L. A Shodex RI-201 refractive index detector was used at a temperature of 35 °C. Data was acquired with Polymer Laboratories Cirrus version 2.0, and 16 narrow polystyrene standards from Polymer Laboratories, fitting to a third-order polynomial curve over the range of 2750–0.580 kg/mol, were applied to calibrate the system.

For NMR analysis, PMMA and PS samples were dissolved in deuterated chloroform (CDCl_3). PEO and egg white protein

samples were dissolved in deuterated water (D_2O). Typically, ~10–15 mg of sample was accurately weighed on a balance (accurate to 0.1 mg) to prepare a concentrated stock solution, e.g., 1 wt %. An aliquot of the stock solution was then transferred and weighed into a glass vial and further diluted to the target concentration by adding a specific amount of solvent with 1 mM chromium(III) acetylacetonate (as the relaxation agent). The dilution level was further cross checked spectroscopically for consistency. For PMMA, 825 ± 5 mg of the above-diluted solution was consistently added into a 5 mm NMR tube. Similarly, 590 ± 10 mg of each PEO or protein sample solution was consistently added into a 5 mm NMR tube. All of the samples were freshly prepared and equilibrated at 25 °C for 10 min prior to any diffusion measurement. Solvent evaporation was negligible for most of the experiments (<1 h) and the NMR tube was sealed when experimental time lasted longer than 5 h. ^1H NMR diffusion measurements were conducted on a 400 MHz Bruker Avance spectrometer equipped with a broad band 5 mm probe. The probe temperature was calibrated and controlled at 25 ± 0.1 °C. Gradient calibration was performed by measuring diffusion of residual $^1\text{H}_2\text{O}$ in D_2O to yield a self-diffusion coefficient of $1.90 \times 10^{-9} \text{ m}^2/\text{s}$.²⁶ Pulsed gradient double spin echo (PGDSE)²⁷ was employed to measure PMMA and PEO diffusion successively by monitoring the variation of the $I_{-\text{OCH}_3}$ and $I_{-\text{OCH}_2}$ signal as a function of magnetic field gradient g . Pulsed gradient double stimulated echo (PGDSTE)²⁸ was employed to measure PS and protein diffusion successively. Around 4–4800 scans were recorded to generate adequate spectra sensitivity with a total repetition time being 4–5 $\times$ of the longitudinal relaxation time T_1 . For polymer samples, a distribution of diffusion coefficients often exists due to the finite polydispersity⁴ and leads to a non-Gaussian NMR signal

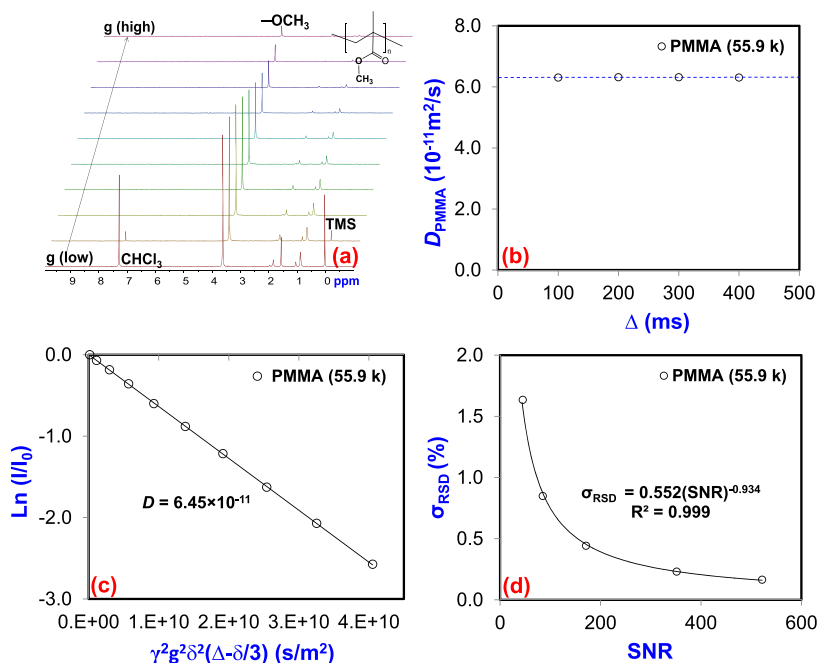


Figure 1. (a) Attenuation of $-\text{OCH}_3$ signal (PMMA) in ^1H NMR diffusion spectra as a function of gradient strength (g) at 25 °C. (b) Independent D_{PMMA} on diffusion time (Δ) confirms the effective suppression of thermal convection for diffusion measurement in CDCl_3 . (c) Representative plot of $\ln(I/I_0)$ vs $\gamma^2 g^2 \delta^2 (\Delta - \delta/3)$ to determine D_{PMMA} of a monodisperse sample ($M = 55.9\text{ k}$). (d) Plot of relative standard deviation σ_{RSD} vs the SNR of $-\text{OCH}_3$ peak with $g \sim 0$. σ_{RSD} characterizes the precision of measurement on D_{PMMA} , which is calculated by averaging D_{PMMA} over 20 replicates.

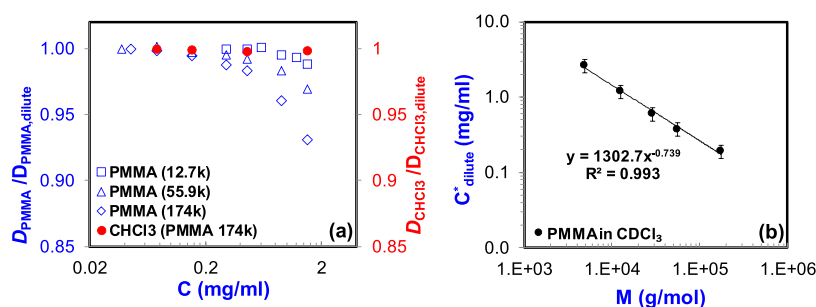


Figure 2. (a) Plot of $D_{\text{PMMA}}/D_{\text{PMMA,dilute}}$ vs polymer concentration (C) at different M and $D_{\text{CHCl}_3}/D_{\text{CHCl}_3,dilute}$ vs C at 25 °C in CDCl_3 . (b) $\log C^*_{\text{dilute}} - \log M$ plot for PMMA in CDCl_3 . The slope corresponds to $\alpha = 0.58$ using eq 4 and it agrees well with the value ($\alpha = 0.56$) based on $\log D - \log M$ plot under the sufficiently dilute condition (Figure 3).

decay in the diffusion measurement. The following equation applies to describe the NMR signal attenuation in the presence of a distribution of diffusion coefficients

$$I = \int P(D) \exp\left(-D\gamma^2\delta^2g^2\left(\Delta - \frac{\delta}{3}\right)\right) dD \quad (5)$$

D is the self-diffusion coefficient, γ is the gyromagnetic ratio of nuclei, δ is the effective gradient pulse duration, and Δ is the diffusion time. In this study, δ and Δ were set as 6 and 100–200 ms, respectively. For monodisperse samples, g varied linearly in the range of 2.5–48 G/cm in 10 steps to result in at least 80% of signal attenuation. For the polydisperse PS sample, g varied linearly from 2.5 to 48 G/cm in 16 steps to result in $\sim 95\%$ of signal attenuation, so there are enough points in both low and high-gradient regions to reflect the distribution width. More scans are often needed for a polydisperse sample to enhance spectra quality for curve fitting, especially in the high-gradient region where spectra tend to be noisy due to signal attenuation. I is the NMR signal intensity and $P(D)$ represents the distribution of diffusion coefficients. The general format of $P(D)$ is unknown and one good approximation commonly used is the log-normal distribution⁴

$$P(D) = \frac{1}{\sigma D \sqrt{2\pi}} \exp\left[-\frac{(\log[D] - \log[D_0])^2}{2\sigma^2}\right] \quad (6)$$

D_0 and σ are parameters that characterize the distribution function. $P(D)$ can be determined by fitting D_0 and σ based on the signal decay described by eq 5 and then converted into the mass-weighted molecular weight distribution $P(M)$ via $D = kM^{-\alpha}$ and the following equation⁴

$$|P(D)dD| = |P(M)dM| \quad (7)$$

RESULTS AND DISCUSSION

Evaluation of eq 4 requires reliable determination of C^*_{dilute} , which necessitates high-quality diffusion data. We design the experiment by selecting monodisperse poly(methyl methacrylate) (PMMA) as a surrogate to polystyrene (PS) standards due to its similarity to PS in many aspects²⁹ and the lack of homonuclear coupling interference on the $-\text{OCH}_3$ group along the polymer backbone. This allowed utilizing the double spin echo²⁷ for diffusion measurement to effectively suppress thermal convection without compromising the spectra sensitivity. In theory, spectra sensitivity using double spin echo is 4× higher than a double stimulated echo. Due to the

relaxation difference between longitudinal relaxation T_1 and transverse relaxation T_2 , $\sim 3.3\times$ of sensitivity gain is achieved for PMMA using double spin echo as compared to the double stimulated echo,²⁸ which leads to a significant reduction of experimental time ($\sim 10\times$) and it represents a crucial step for efficient and reliable analysis. Figure 1a representatively shows diffusion spectra of PMMA in deuterated chloroform (CDCl_3) vs variation of gradient strength, g . The characteristic $-\text{OCH}_3$ peak of PMMA and solvent peaks are labeled accordingly. The measured D_{PMMA} is free of artifact, which is confirmed by its independence on diffusion time Δ and the excellent fit shown in Figure 1b,c, respectively. Chloroform diffusion (D_{CHCl_3}) is measured separately using lower gradient strengths. For PMMA, spectra quality is characterized by the signal-to-noise ratio (SNR) of $-\text{OCH}_3$ peak in the diffusion spectrum when $g \sim 0$. For a given SNR, the relative standard deviation σ_{RSD} (SI, eq S11) is calculated by averaging 20 replicates, and the relationship between σ_{RSD} and SNR is well characterized by the power law ($R^2 = 0.999$), as shown in Figure 1d. The result is consistently leveraged to all D_{PMMA} measurements with SNR ~ 350 being selected ($\sigma_{\text{RSD}} < 0.25\%$) as the experimental criteria so that high-precision D_{PMMA} can be measured down to the concentration level ~ 40 ppm in 3.5 h.

Diffusion of PMMA (D_{PMMA}) and residual protonated chloroform (D_{CHCl_3}) in deuterated chloroform (CDCl_3) are normalized with respect to their corresponding values under the sufficiently dilute condition ($\frac{dD_{\text{PMMA}}}{dC} = 0$) and represented by $D_{\text{PMMA}}/D_{\text{PMMA,dilute}}$ and $D_{\text{CHCl}_3}/D_{\text{CHCl}_3,dilute}$. Figure 2a exhibits $D_{\text{PMMA}}/D_{\text{PMMA,dilute}}$ vs polymer concentration (C) at three different M . $D_{\text{CHCl}_3}/D_{\text{CHCl}_3,dilute}$ is also plotted vs C for one PMMA sample ($M = 174\text{k}$). $D_{\text{PMMA}}/D_{\text{PMMA,dilute}}$ shows various levels of dependence on C , whereas no variation of $D_{\text{CHCl}_3}/D_{\text{CHCl}_3,dilute}$ is observed in the same concentration range. The different responses between D_{PMMA} and D_{CHCl_3} to the variation of C counterintuitively indicate that the change in viscosity with C is not uniform between a polymer and solvent if one simply attributes the change of diffusion to the concentration (or volume) effect based on Einstein's viscosity theory.^{30,31} The Stokes–Einstein equation³² (eq 8) which describes the Brownian motion of a macroscopic entity in a continuum media, e.g., polymer diffusion in a solvent, can be further used to understand the phenomenon. For polymer chains, the hydrodynamic radius R_h is an effective term, which depends on the shape and interactions. Since the temperature (T) is constant and media viscosity (η) remains unchanged as

indicated by D_{CHCl_3} , the dependence of $D_{\text{PMMA}}/D_{\text{PMMA,dilute}}$ on C simply indicates that polymer chain interactions still exist even under low concentrations.

$$D = \frac{k_B T}{6\pi\eta R_H} \quad (8)$$

In some studies,^{11,33} independence of solvent diffusion on C has been regarded as a sign of the sufficiently dilute condition (negligible polymer interactions). The result illustrated in Figure 2a clearly contradicts such a consideration, demonstrating that solvent diffusion is inadequate for such a purpose. Even at a moderate $M = 174\text{k}$, D_{PMMA} does not become invariant until C falls below 0.2 mg/mL , which is lower than most of the concentrations using NMR diffusion reported up to date^{4–6,8–22} and still not adequate for samples with higher M . C_{dilute}^* is determined with high confidence (SI) and plotted against M using the double logarithm mode to yield a linear regression in Figure 2b. The slope yields $\alpha = 0.58$ based on eq 4, which agrees well with that determined by the $\log D$ – $\log M$ plot under the dilute condition ($\alpha = 0.56$, Figure 3), thereby

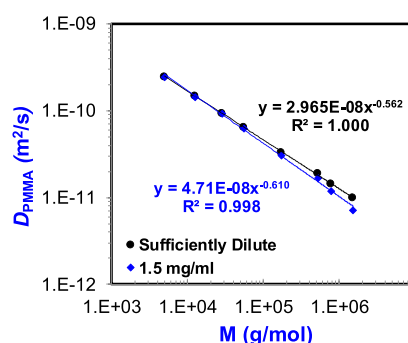


Figure 3. Plot of D_{PMMA} vs M is compared under two conditions in CDCl_3 : 1.5 mg/mL ($0.1\text{ wt } \%$) and the sufficiently dilute concentration, which is predicted based on Figure 2b. Though linear regression results in both cases with $R^2 \sim 1$, the obtained slope (0.56 vs 0.61) is quite different from each other, indicating 1.5 mg/mL is not sufficiently dilute.

providing solid evidence to verify eq 4. We also observed the same pattern for poly(ethylene oxide) (PEO) in D_2O (SI, Figure S1) that further strengthens the validity of eq 4. By approximating MWD with the log-normal distribution (SI, eqs S4–S9), eq 9 links C_{dilute}^* to some key polymer characteristics, including chain conformation (α), characteristic molecular weight M_0 , and PDI

$$C_{\text{dilute}}^* = C_0' M_0^{1-3\alpha} \text{PDI}^{-(3\alpha-1)^2/2} \quad (9)$$

C_0' is a constant, $M_0 = \frac{M_w}{\sqrt{\text{PDI}}}$, and M_w is the weight average M . A rough comparison reveals that C_{dilute}^* for PMMA is consistently $\sim 100\times$ lower than C^* of polystyrene in a good solvent.²⁴ Such a remarkable gap between C^* and C_{dilute}^* should remind the audience why the term “below C^* ” or “significantly below C^* ” can be ambiguous to specify the sufficiently dilute condition. This also provides a rule of thumb to estimate C_{dilute}^* via C^* , which is often easy to determine or estimate via $C^* = \frac{M}{N_A R_g^3}$.²⁴ R_g ($\sim R_h$) is the radius of gyration. Using this equation, the estimated C^* is $\sim 200\times$ higher than C_{dilute}^* for PMMA, which is also consistent with the order of magnitude difference of C^* and C_{dilute}^* between PMMA and PS. The

finding further relates to polymer characterization using other techniques, such as GPC and rheology, by which the flow-induced shear force will cause chain deformation (increase in α) and it may necessitate a significantly lower C_{dilute}^* to avoid chain interactions at high flow rates (i.e., large shear force).

Measurement of D_{polymer} beyond the sufficiently dilute condition will consequently induce systematic errors in M /MWD characterization, which can be particularly problematic for samples with high M . For monodisperse samples, Figure 3 compares D_{PMMA} vs M in CDCl_3 under two conditions: 1.5 mg/mL ($0.1\text{ wt } \%$) and sufficiently dilute condition with M in the range from 4.9 to 1520k . For $M > 200\text{k}$, the sufficiently dilute concentration is projected based on the result obtained from Figure 2b. Although the violation of the sufficiently dilute condition will lead to a deviation from linear regression in the $\log D$ – $\log M$ plot, such a deviation actually occurs smoothly as M increases, which often makes it hard to be perceived by visual inspection. Only the data point at $M = 1520\text{k}$ (1.5 mg/mL) appears to be somewhat off the fitting curve that might shed some light on the potential problem. Without prior knowledge of eq 4, it is difficult to recognize the underlying problem merely based on the R^2 value. In other words, one can easily get different fitting curves with a high quality (R^2 value close to 1) in $\log D$ – $\log M$ plot even using the same set of polymer standard. This is likely another one of the main reasons that disparity-associated D – M relationships widely exists in the literature. Using monodisperse PMMA as an example, quantitative analysis shows that α and k will increase by 8.6 and 59% , respectively, as the condition transits from sufficiently dilute to 1.5 mg/mL , corresponding to an increase of error in M from 2.5 to 28% as M increases from 20 to 1000k . This clearly shows the potential level of error associated with M (via diffusion analysis) if sufficiently dilute condition criterion is not satisfied. Taking one of the examples mentioned in the introduction, the difference in α values reported by Arrabal-Campos et al. and Li et al. for the same system can be just rationalized based on the above analysis. Based on Figure 3, one can easily tell the condition used by Arrabal-Campos et al. (0.9 mg/mL for PS samples with M up to 1 million) significantly violated the sufficiently dilute condition in the high M region ($M > 200\text{k}$). The corresponding polymer diffusion therefore will appear to be much slower. Including these data points in the D – M calibration curve will tilt up the linear regression in the $\log D$ – $\log M$ plot to artificially inflate k and α values. Consequently, the claimed universal calibration likely remains invalid since it depends on sample concentration. In contrast, the work by Li et al. satisfied the sufficiently dilute condition given the studied M range ($<200\text{k}$). The determined α value also remains highly consistent regardless of the region of M data is used, which demonstrates that there is no change in chain conformation and quality of diffusion measurement should be very high and the uncertainty associated with α should be $\sim 1\%$ or less, provided sufficient spectra sensitivity is viable.

For polydisperse samples, Figure 4 compares MWD of the PS1683 sample based on PMMA ($\alpha = 0.562$ and $k = 2.965 \times 10^{-8}$) and the work by Guo et al.²² ($\alpha = 0.52$ and $k = 2.19 \times 10^{-8}$). The result obtained near the sufficiently dilute condition ($0.01\text{ wt } \%$ PS in CDCl_3) shows a much better agreement with the GPC analysis than that of Guo et al., which utilized $0.3\text{ wt } \%$ PS in CDCl_3 for a broad range of M . The determined M_n and M_w are also summarized in Table 2 for more specific

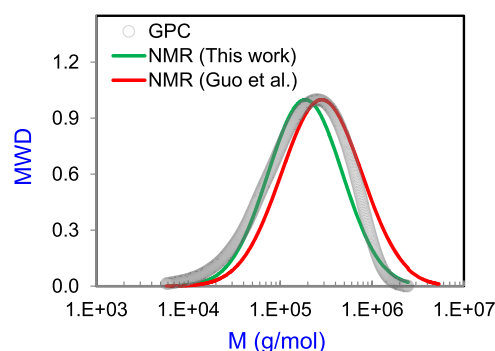


Figure 4. MWD of a polydisperse PS sample determined by GPC and NMR. NMR analysis based on the current work exhibits a much better agreement with GPC as compared to that by Guo et al., which largely deviates from the sufficiently dilute condition.

Table 2. M_n and M_w of the Polydisperse PS1683 Sample Determined by Different Methods

M/method	NMR _{Guo et al.}	NMR _{this work}	GPC
M_n (kg/mol)	164 ± 3	118 ± 3	102
M_w (kg/mol)	442 ± 35	277 ± 22	275

comparison. One should note that the confidence level of these results is worse than that of PMMA since the spectra sensitivity of PS is not adequate at such a low concentration. Even with >4500 scans, it is still very challenging to enhance spectra quality in the high-gradient region, which weighs toward the determined distribution width and M_w . As a result, the error for M_w is higher than that of M_n . The above discussion aims to help the general audience understand the risk of going beyond sufficiently dilute condition for MWD characterization by NMR diffusion. We also suggest the use of sufficiently dilute condition as one possible unified criterion so that comparison of polymer diffusion and MWD is viable and more straightforward among different labs and polymer systems.

Validation of eq 4 not only establishes the basis to determine C_{dilute}^* for polymer characterization but holds two important applications for polymer characterization by diffusion. First, D_{polymer} will be insensitive to C variation within a certain range if polymer chains adopt the globular structure, i.e., $\alpha = 1/3$. Under this circumstance, polymer chains become highly folded that drastically diminish the impact of chain interactions on D_{polymer} and the dependence of C_{dilute}^* on M will vanish. Many proteins are well known to form the globular structure, and we use albumin of egg white ($M \sim 43\text{k}$) to examine the above prediction. Figure 5 compares diffusion distribution $P(D)$ between egg albumin protein and monodisperse PEO standard ($M = 32.5\text{k}$) in D_2O . $P(D)$ remains invariant with protein concentration, in contrast to the large variation of $P(D)$ with PEO concentration, which verifies the prediction stated above. This further provides a new avenue to evaluate the folding of protein in solution, which is an important topic in bioscience.

Second, the validation of eq 4 confirms $f_{v,\text{dilute}}^*$ as a constant for a given system, thereby opening a new avenue toward characterizing chain architecture and M . By approximating $P(D)$ with the log-normal distribution, derivation of eq 2 yields the following (SI, eqs S8–S10)

$$f_{v,\text{dilute}}^* = N_A C_{\text{dilute}}^* \frac{\varepsilon_0 R_{h,0}^3}{M_0} \text{PDI}^{(3\alpha-1)^2/2} \quad (10)$$

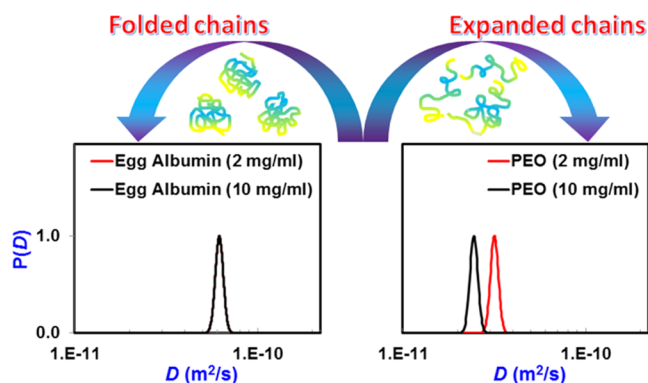


Figure 5. Diffusion distribution $P(D)$ of a globular protein—egg albumin and monodisperse PEO ($M = 32.5\text{k}$) in D_2O . $P(D)$ shows no dependence on concentration for protein but a significant dependence on concentration for PEO.

$\varepsilon_0 R_{h,0}^3$ represents the volume (v_p) pervaded by individual polymer chain with ε_0 introduced as a prefactor, e.g., $\varepsilon_0 = 4/3\pi$ for spherical conformations. Considering chain conformation as the major factor that governs ε_0 , $f_{v,\text{dilute}}^*$ can be approximated as a constant for linear polymers with a similar chain conformation in the same solvent. PMMA can be treated as a PEO analog with short branches, meaning the determined M of PEO via eq 10 will be significantly lower than that based on size, i.e., PMMA apparent M (M_{ap}). For simplicity, we set ε_0 to 1. Diffusion analysis of PMMA standards and Figure 2b yield $f_{v,\text{dilute}}^*, \text{CDCl}_3 = 0.0012 \pm 0.00018$, which in combination with eq 10 allows determination of M (M_0) of two monodisperse PEO samples in CDCl_3 : PEO8.5k and PEO32.5k. Table 3 compares

Table 3. Molecular Weight of Monodisperse PEO Samples Determined by Different NMR Methods

sample	M_{ap} (kg/mol)	M_0 (kg/mol)
PEO8.5k	16.5 ± 0.2	8.5 ± 1.7
PEO32.5k	63.2 ± 0.6	36.5 ± 7.3

the results of M_0 and M_{ap} for PEO based on PMMA. In summary, M_{ap} is significantly higher than M_0 , indicating PEO has much fewer branches than PMMA. In addition, M_0 determined by eq 10 is much closer to the actual M as expected.

CONCLUSIONS

We have demonstrated the critical role of C_{dilute}^* to specify the sufficiently dilute condition for M and MWD characterization and further linked it to chain conformation, M and PDI. Polymer and solvent diffusion respond differently to concentration variation, highlighting the unique role of chain interactions even at concentrations significantly below C^* . Quantitative analysis indicates that D_{polymer} measurement above the sufficiently dilute condition would lead to a systematic and potentially large error in M and MWD analysis. These findings help to understand the data disparity in the literature and provide useful guidance toward M /MWD characterization by NMR diffusion measurements. The study offers key insights that lead to the discovery of the invariant macromolecular diffusion on concentration for globular structure and provides a different approach for characterizing polymer molecular weight and chain architecture.

■ ASSOCIATED CONTENT

■ Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.analchem.1c00793>.

Derivation of C_{dilute}^* and $f_{v,\text{dilute}}^*$, determination of C_{dilute}^* and confidence level, plot of PEO diffusion (D_{PEO}), and C_{dilute}^* vs M in D_2O (PDF)

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Notes

The authors declare no competing financial interest.

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